

Particle Accelerators:

A Particle accelerator is a device for increasing the kinetic energy of electrically charged particles. It is also a machine that uses electromagnetic fields to propel charged particles to very high speeds and energies and to contain them in well defined beams. The elementary particles such as electrons and protons are accelerated to a very high energy. There are more than 30,000 Particle accelerators around the world. Particle accelerators, like the Large Hadron Collider (LHC), shoots particles at incredibly high speeds, smash them together and observe the fallout like Higgs Boson, also called God particle.

Types of Particle accelerators:

* On the basis of the path followed by the accelerating particles, there are two types

i) Linear accelerators: Particles propel in straight line.

ii) Circular accelerators: Particles propel in circular track.

* The methods of acceleration can be classified into three groups:

i) Direct field: A particle passes only once through an electric field with a high p.d. set up by electrostatic generators.
e.g. linear accelerator

ii) Inductive field: The particle successively passes through increasing magnetic field which inductively couples power into the charged particle beam.

e.g. Betatron

iii) Resonance field: In magnetic resonance accelerators, the particles being accelerated repeatedly passes through an alternating electric field along a closed path. Its energy being increased each time.

e.g. cyclotron.

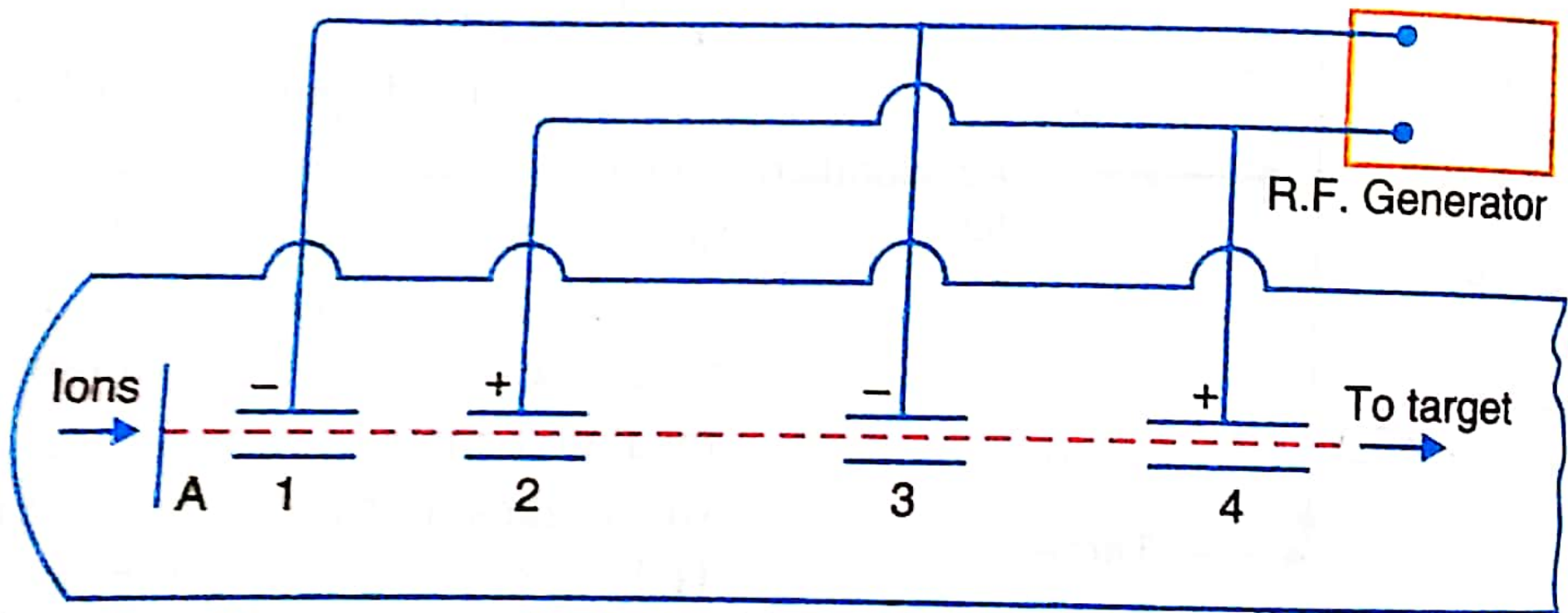
The everyday examples of particle accelerators are Cathode ray tubes found in television sets and X-ray generators. These low energy accelerators use a single pair of electrodes with a dc voltage of a few thousand volts between them.

In an accelerator, the particle source provides the particles such as protons & electrons. The particle accelerators use electric fields to speed up and increase the energy of a beam of particles, which are steered and focused by magnetic fields.

The Linear Accelerator:

By the application of potential difference of millions of volts, direct acceleration of particles is a difficult job, due to insulation difficulties. For such high energies, acceleration of the particles is achieved in small successive steps. In such machines, the P.d. between different parts of the machine & between the machine & the earth, is maintained low, compared with P.d. corresponding to the ultimate energy acquired by the particles. The device linear accelerator is employing the same method. In this machine, high energy particles are produced without employing high P.d. by using the principle of synchronous acceleration.

The figure shows the schematic diagram of a linear accelerator. It consists of a series of coaxial hollow metal cylinders 1, 2, 3, 4... etc. They are arranged linearly in a glass vacuum chamber. The alternate cylinders are connected together, the odd numbered cylinders being joined to one terminal and the even numbered ones to the second terminal of high frequency (H.F.) oscillator. Thus in one-half cycle, if tubes 1 & 3 are positive, tube 2 & 4 will be negative. After half a cycle, the polarities are reversed i.e. 1 & 3 will be negative & 2 & 4 positive. The ions are accelerated only in the gap between the tubes where they are acted upon by the electric field present in the gaps. The ions travel with constant



Velocity in the field free space inside the drift tubes.

The positive ions enter along the axis of the accelerator from an ion source through an aperture A. Suppose a positive ion leaves A & is accelerated during the half cycle, when the drift tube 1 is negative with respect to A. Let 'e' be the charge & m the mass of the ion, as well as V be the Potential of drift tube 1 with respect to A. Then velocity V_1 of the ion on reaching the drift tube is given by

$$\frac{1}{2} m V_1^2 = eV \quad \text{or} \quad V_1 = \sqrt{\frac{2eV}{m}}$$

The length of the tube 1 is so adjusted that as the positive ions come out of it, the tube has a positive potential & the next tube 2 has a negative Potential. The positive ion is again accelerated in the space between the tubes 1 & 2. On reaching the tube 2, the velocity V_2 is,

$$\frac{1}{2} m V_2^2 = 2eV \quad \text{or} \quad V_2 = \sqrt{2} \sqrt{\frac{2eV}{m}}$$

$$\therefore V_2 = \sqrt{2} V_1$$

on the other hand,

the ion, on coming out of tube 2, may find tube 3 just negative & the tube 2 positive, it must take the same time to travel through the tube 2. Since $V_2 = \sqrt{2} V_1$, the length of tube 2 must be $\sqrt{2}$ times the length of tube 1. For successive acceleration

in successive gaps the tubes 1, 2, 3 etc must have lengths proportional to $1, \sqrt{2}, \sqrt{3} \dots$ etc.

$$\text{i.e. } l_1 : l_2 : l_3 \dots \text{etc} = 1 : \sqrt{2} : \sqrt{3} \dots \text{etc.}$$

Energy Carried by the ion:

Let n be the number of gaps that the ion travels in the accelerator and v_n = final velocity acquired by the ion after travelling n gaps, in between drift tubes.

Then velocity acquired by the emerging positive ion on emerging out of n th tube

$$= \sqrt{n} \sqrt{\frac{2eV}{m}}$$

The Kinetic energy acquired by the ion

$$K.E = \frac{1}{2} m v_n^2 = \frac{1}{2} m \left(\sqrt{n} \sqrt{\frac{2eV}{m}} \right)^2$$

$$\boxed{K.E = Ven}$$

This is energy relation of a particle in linear accelerator. The relation shows that the energy of accelerating particle depends upon -

(i), the total number of gaps (n)

(ii), the energy gained in each gap (eV).

Limitations: (i) The length of the linear accelerator becomes inconveniently large (ii) Difficult to maintain vacuum in a large chamber (iii) The ions are injected in the form of pulse.

The Cyclotron:

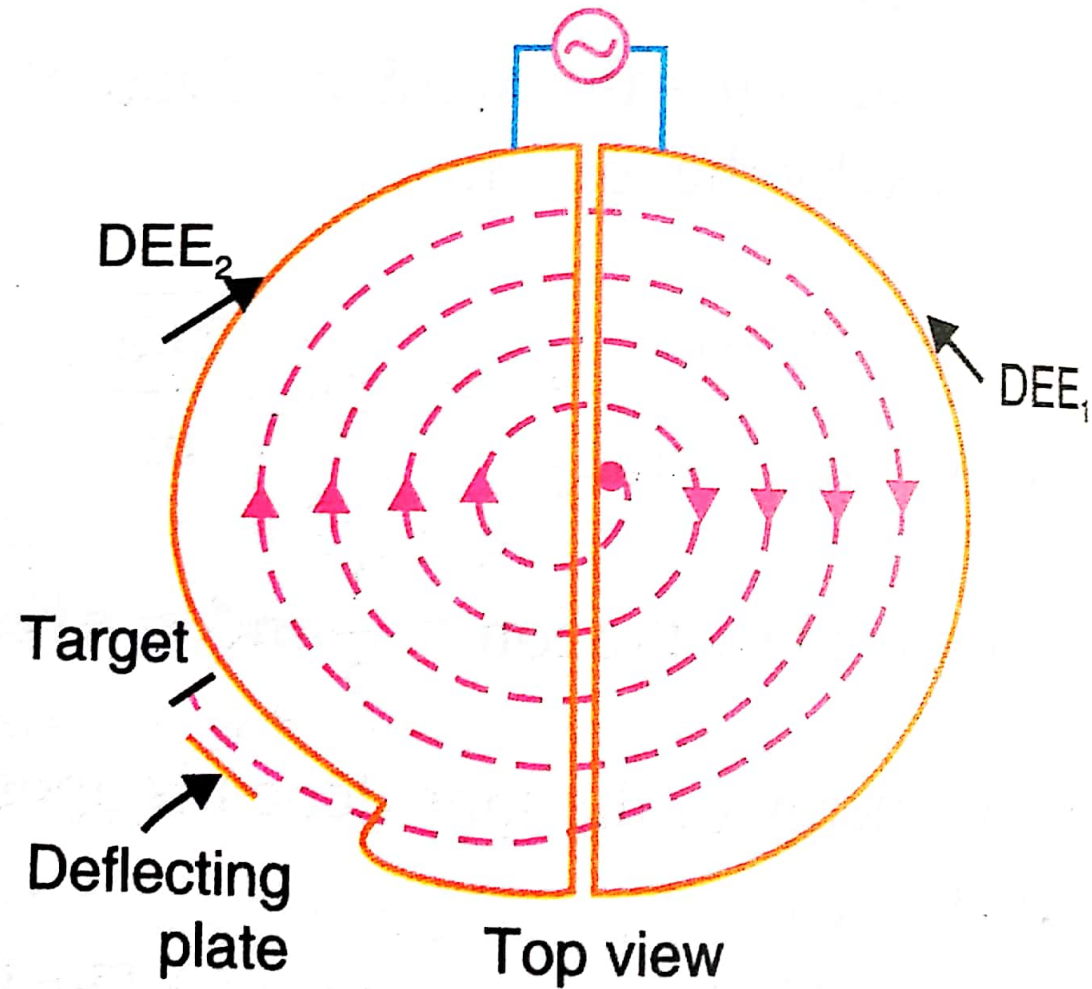
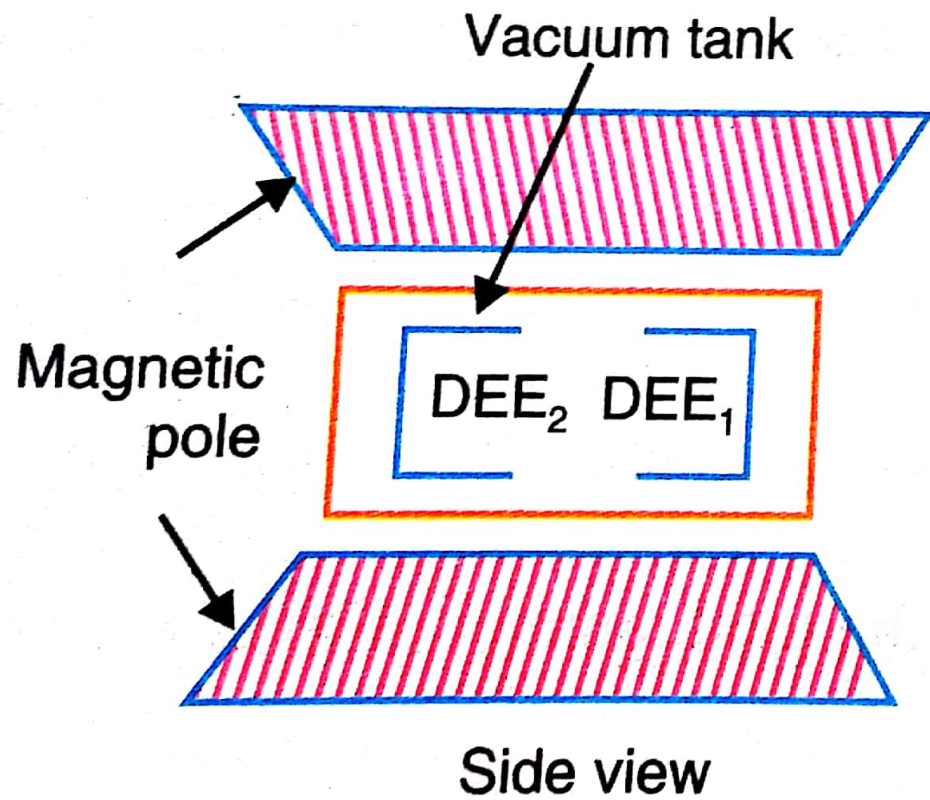
A Cyclotron is a type of Particle accelerator invented by Ernest O. Lawrence in 1929-1930. A cyclotron accelerates charged particles outward from the centre of a flat cylindrical vacuum chamber along a spiral path. It is an electrically powered machine which produces a beam of charged particles that can be used for medical, industrial and research process.

Construction:

The cyclotron consists of two hollow semicircular metal boxes D_1 and D_2 called 'dees', these are insulated from each other and are enclosed in another vacuum chamber. The dees are connected to a powerful radio frequency oscillator. A source of ion is located near the mid point of the gap between the dees. The whole apparatus is placed between the pole pieces of a strong electromagnet. The magnetic field is perpendicular to the plane of the dees.

Theory:

Suppose dees D_1 & D_2 are at negative and positive A.C. potential respectively. The positive ion lies at the centre of two dees. Initially it moves towards D_1 and enter into it. The ion gets accelerated in the gap and enter into the dee D_1 with velocity U . If potential



difference between the dees is V , then from principle of conservation of energy

$$eV = \frac{1}{2}mv^2$$

where 'e' is charged of the particle & 'm' its mass. As long as the particle remains inside the dee it is not accelerated, since it is field free region. The magnetic field is applied perpendicular to the plane of motion of the particle, it urges particle to travel in circular path of radius 'r'.

$$\therefore Bev = \frac{mv^2}{r} \quad \text{or} \quad r = \frac{mv}{Be} \quad \text{--- (1)}$$

where B is strength of applied magnetic field.

Also from eqⁿ (1) $v = \frac{Ber}{m} \quad \text{--- (2)}$

The angular velocity of the ion in its circular path be ω , so $\omega = \frac{v}{r} = \frac{Ber}{m} \cdot \frac{1}{r}$

$$\omega = \frac{Be}{m} \quad \text{--- (3)}$$

Let 't' be the time taken by the ion to travel the semicircular path, then

$$t = \frac{\pi}{\omega}$$

$$\therefore t = \frac{\pi}{\frac{Be}{m}} = \frac{\pi m}{Be} \quad \text{--- (4)}$$

$$\begin{cases} \omega = 2\pi f - \text{circle} \\ \omega = \pi f - \text{Semicircle} \\ \therefore \omega = \frac{\pi}{t} \\ t = \frac{\pi}{\omega} = \frac{\pi}{\omega} \end{cases}$$

f = frequency of oscillator

The value of B & f are adjusted that, by the time the ion has described a semicircular path. Now the ion enters into

the space between two dees and at the same time polarity (ies) of the applied ac in D_1 & D_2 reverses. D_2 has become negative with respect to D_1 . The ion is then accelerated towards D_2 & enters the space inside it with a greater velocity. The ion describes semicircle of greater radius in the dee. From eqn (4), the time 't' is independent of both 'r' and 'v'. Hence, the ion completes the semicircular path in the same time, whatever may be its radius. On doing so time and again, the ion reaches the periphery of the dee with possible greatest velocity. The ion spirals round in circles of increasing radius & acquires high energy. Finally the ion exits out through the window as indicated in figure.

Energy of an ion:

Let $r_{\max}$ & $v_{\max}$ be the radius of the outermost orbit described by the particle and the maximum velocity gained by the ion, respectively. Then in equilibrium condition, centripetal force is balanced by magnetic force, as

$$Bev_{\max} = \frac{mv_{\max}^2}{r_{\max}}$$

$$\text{or } v_{\max} = B \frac{e}{m} r_{\max} \quad \dots (5)$$

The energy of the ion

$$E = \frac{1}{2} mv_{\max}^2 = \frac{1}{2} m \left(B \frac{e}{m} r_{\max} \right)^2$$

Inserting $B = 1.5 \text{ T}$ and $r_{\max} = 0.5 \text{ m}$ in the above eqn.

$$E = \frac{B^2 r_{\max}^2}{2} \left(\frac{e^2}{m} \right) \quad \dots (6)$$

The condition for acceleration of the ion in the gap between the dees is

$$\left. \begin{array}{l} \text{The time taken by} \\ \text{the ion to travel the} \\ \text{Semicircular Path} \end{array} \right\} = \left\{ \begin{array}{l} \text{Half the time period} \\ \text{of oscillation of the} \\ \text{applied high frequency voltage} \end{array} \right.$$

If T is the time period of oscillation, then $T/2$ is half time period. So,

$$\frac{\pi m}{Be} = \frac{T}{2} \quad \text{or} \quad T = \frac{2\pi m}{Be}$$

Then frequency of oscillation $f = \frac{1}{T}$.

$$\text{or } f = \frac{Be}{2\pi m} \quad \text{or } Be = 2\pi m f \quad \text{--- (7)}$$

Hence the energy of the ion/charged particle is given by

$$E = \frac{B^2 r_{\max}^2}{2} \left(\frac{e}{m} \right)^2$$

$$= (Be)^2 \cdot \frac{1}{2} \cdot \frac{r_{\max}^2}{m}$$

$$= (2\pi m f)^2 \cdot \frac{1}{2} \cdot \frac{r_{\max}^2}{m} \quad [\text{from eqn (7)}]$$

$$E = 2\pi^2 r_{\max}^2 f^2 m \quad \text{--- (8)}$$

The eqn (8) expresses maximum possible energy of the particle when comes out from the outermost orbit of the Circular/Spiral path inside the dee.

Limitations of the cyclotron: As we see, there is influence of mass of the particle in the energy

relation in eqn (8), there should be relativistic correction in the mass. When the particle moving with velocity v , then mass in relativistic approach is $m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$, where $m_0 = \text{rest mass}$
 $c = \text{velocity of light}$

From eqn (4) $t = \frac{2\pi m}{Be} = \frac{T}{2} \Rightarrow T = \frac{2\pi m}{Be}$

Then frequency of the ion $f = \frac{1}{T}$.

or $f = \frac{Be}{2\pi m} = \frac{Be}{2\pi} \frac{\sqrt{1 - v^2/c^2}}{m_0}$... (9)

The relation (9) shows that as velocity of the particle increases the frequency of rotation decreases. Thus the ion lags behind the applied potential and finally the particle is not accelerated further. The limitations can be overcome in the following two ways.

(i). Field Variation: (Synchrotron).

We have $f = \frac{Be}{2\pi} \frac{\sqrt{1 - v^2/c^2}}{m_0}$ with the increase in velocity of the particle, if magnitude of magnetic field is increased in the same way such that the whole product in above relation remains constant, then f will be constant. This type of machine in which frequency of electric field is kept constant & magnetic field is varied is called synchrotron.

(ii). Frequency Modulation: (Synchro-cyclotron). In this case, the frequency of applied a.c. is varied so that it is always equal to the frequency of rotation of ion. This type of machine in which B is kept constant but frequency of applied a.c. is varied is called frequency modulated cyclotron or synchro-cyclotron.

The Betatron:

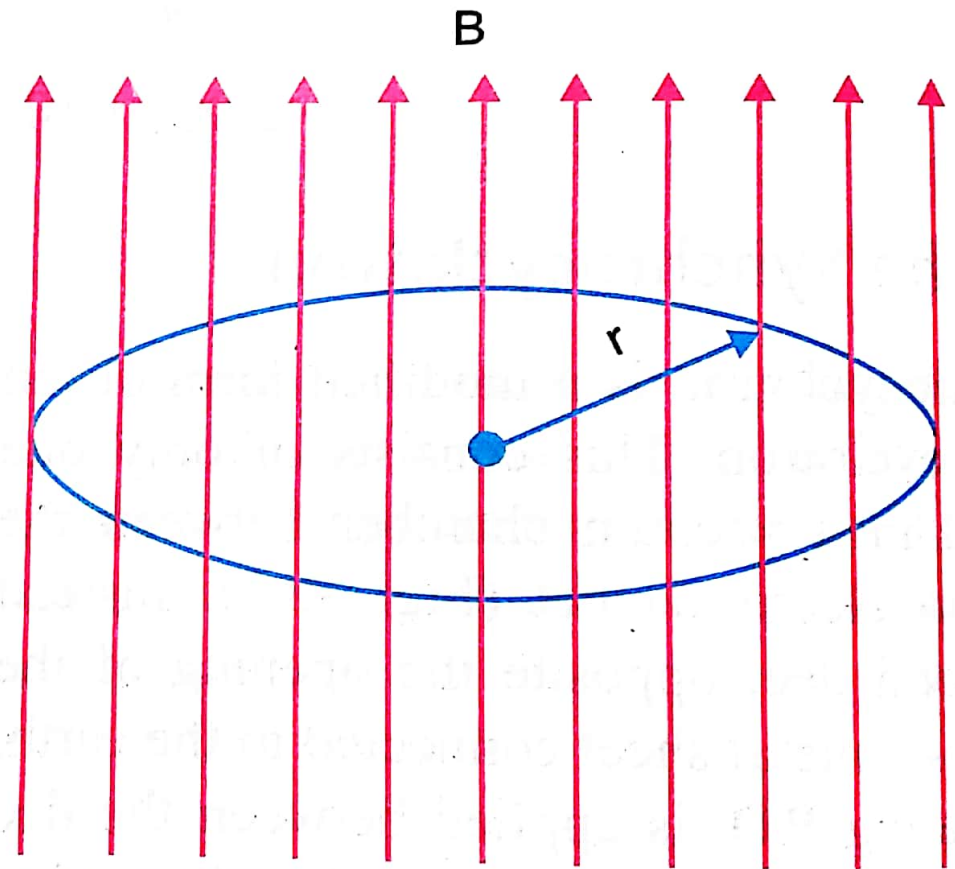
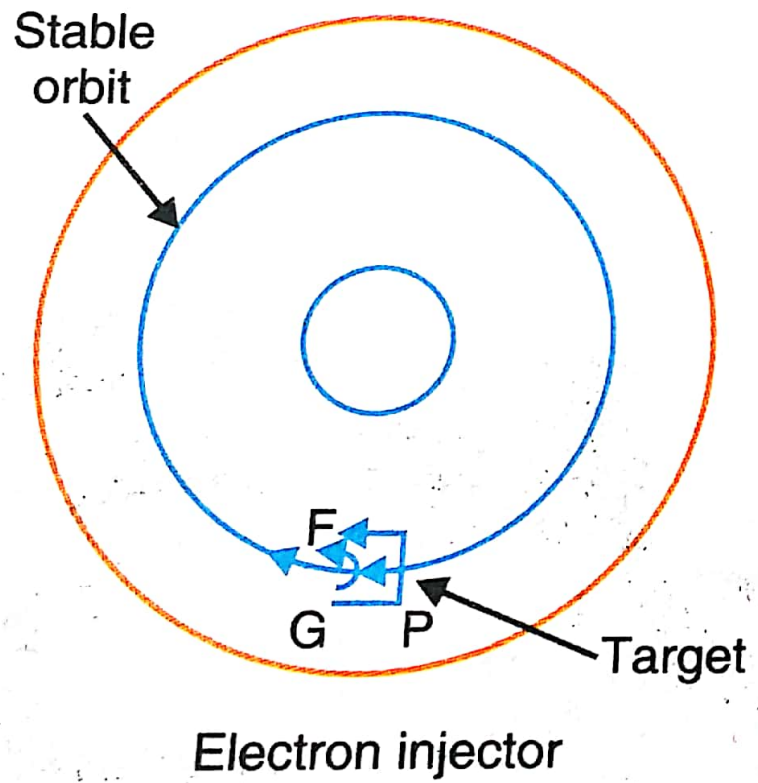
It is a type of particle accelerator that uses the electric field to accelerate electrons (β -particles) to high speeds in a circular orbit. The first successful betatron was completed in 1941 at the University of Illinois by physicist **Donald W. Kerst**. Modern compact betatron designs are used to produce high energy x-rays for varieties of applications. It functions on the principle of a transformer.

Construction:

It consists of a doughnut shaped (torus-shaped) vacuum chamber placed between the pole-pieces of an electromagnet. Alternating current is used in the electromagnet. The electrons are produced by the electron gun and are allowed to move in a circular orbit of constant radius. The magnetic field varies very slowly compared with the frequency of revolution of the electrons in the equilibrium orbit.

The magnetic field acting parallel to the axis of the vacuum tube, produces two effects on the electrons, viz.

- 1) The changing flux due to the electromagnet produces the induced emf which is responsible for the acceleration of the electrons.
- 2) The field of the magnet serves as the



Same time to bend the electrons in a circular path in the chamber and confine them to the region of the changing flux.

Theory:

Suppose the electron is moving in an orbit of radius 'r'. The flux linked with the orbit be Φ . Since there occurs a slight variation in magnetic field, the rate of change of flux linkage be $\frac{d\Phi}{dt}$. As a result the induced emf in the orbit be E . It is expressed as

$$E = - \frac{d\Phi}{dt} \quad \dots (1)$$

The work done on an electron of charge 'e' in one revolution = $Ee = -e \frac{d\Phi}{dt}$ $\dots (2)$

Let F be the tangential electric force acting on the orbiting electron. For one revolution, the path length is $2\pi r$. Then work done on the electron in one revolution = $F \times 2\pi r$ $\dots (3)$

In equilibrium condition, two energy should balance each other, hence from eqⁿ (2) & (3), we get

$$-e \frac{d\Phi}{dt} = F \times 2\pi r$$

$$F = - \frac{e}{2\pi r} \frac{d\Phi}{dt} \quad \dots (4)$$

Due to the force in eqⁿ (4) the velocity of the electron increases, as a result it will try to move into an orbit of larger radius. Due to the

Presence of the magnetic flux perpendicular to the plane of the electron orbit, the electron will experience a radial force inward. It can be expressed as

$$Bev = \frac{mv^2}{r} \quad \text{--- (5)}$$

Where B = magnetic field intensity
 v = Velocity of the electron
 m = mass of electron.

from eqⁿ (4), $mv = Ber$ --- (6)

From Newton's second law of motion,

$$F = \frac{d}{dt}(mv) = \frac{d}{dt}(Ber)$$

$$\text{or } F = er \frac{dB}{dt} \quad \text{--- (7)}$$

To maintain the constant radius of the orbit of the electron, two forces given by equation (4) and (7) should balance each other.

$$\therefore \frac{e}{2\pi r} \frac{d\phi}{dt} = er \frac{dB}{dt}$$

$$\text{or } d\phi = 2\pi r^2 dB$$

Integrating, $\int_0^\phi d\phi = \int_0^B 2\pi r^2 dB$

$$\text{or } \phi = 2\pi r^2 B \quad \text{--- (8)}$$

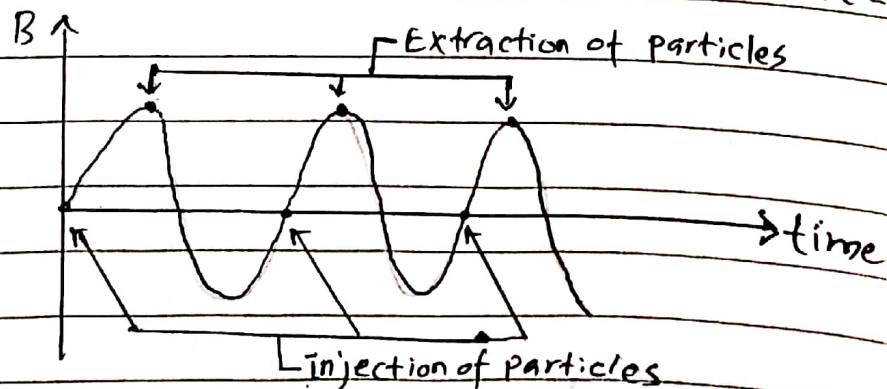
The equation (8) represents the condition under which a betatron works & is called **Betatron**

Condition.

If the uniform magnetic field B acts over an area πr^2 , the magnetic flux linked is $\Phi' = \pi r^2 B$.

The relation (8) shows that, the flux through the orbit is twice the flux enclosed by the orbit, if the magnetic field were to be uniform over the area. It is related to betatron condition.

The variation of magnetic field and injection as well as extraction of particles can be explained with the help of the fig as shown. The electrons are injected into the chamber when magnetic field just begins to rise. The electrons are then accelerated by the increasing magnetic field/flux linked with the electron orbit. During one quarter time period of oscillation the magnetic field reaches its peak value. The electrons get accelerated & make several thousand revolutions. The beam of energised state at the end of $\frac{1}{4}$ time period is extracted and used it to hit the target. After $\frac{1}{4}$ th time period the decrease in flux takes place which decelerate the electrons.



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fig Injection & extraction of particles in betatron.